

Airline Reservation System

Big Data and Data Engineering Midterm — Final Report

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Executive summary

This project implements a small regional-airline reservation system and its first analytical data platform. The transactional application supports direct and one-connection searches, multi-passenger reservations, temporary seat holds, automatic seat assignment, simulated payments, ticket issuance, cancellation, refunds, agency attribution, manifests, and an auditable state history. PostgreSQL row locking and database constraints prevent overselling under concurrent requests.

The application is deployed on AWS using a small EC2 instance and a private Amazon RDS PostgreSQL OLTP database. A second private RDS PostgreSQL instance contains a dimensional OLAP model. An hourly AWS Glue 4.0 Spark job connects to both databases through private JDBC connections, refreshes analytical facts and dimensions, and records reconciliation results. Two Glue crawlers publish 12 selected OLTP tables and 13 OLAP tables in separate Glue Data Catalog databases.

The current list-price run rate is approximately **USD 67.41 per month** before Academy credits and taxes. Changing the refresh frequency from hourly to daily would reduce it to approximately **USD 44.99 per month** without changing the data model.

1. Context, scope, and assumptions

1.1 Business context

The system serves a small regional airline that sells directly and through partner travel agencies. Commercial management needs a reliable transactional system now and data prepared for later analysis of occupancy by route, revenue by fare, and cancellation patterns.

The team did **not receive an additional team-specific business or technical constraint** under section 3 of the assignment. The principal environmental constraint is AWS Academy Learner Lab: IAM creation or modification is prohibited by project policy, only allowed regions/services may be used, and the available budget is limited.

1.2 Explicit sizing assumptions

Assumption	Adopted value
Airline size	8 airports, 12 direct routes, 4 aircraft
Operations	24 flight instances/day
Demand	500 passengers and 250 reservations/day

Seasonal peak	5× normal searches and 3× normal reservation creation
Search horizon	Current time through 180 days
Sales cutoff	60 minutes before first-segment departure
Itinerary	One-way; direct or one connection; maximum 2 segments
Connection	Same airport, 45 minutes to 6 hours
Party size	1–9 passengers, same itinerary and cabin
Cabins	Economy and Business
Currency	COP only; AWS infrastructure costs are stated in USD
Hold	60 minutes pending payment
Quote validity	15 minutes
Cancellation	Whole reservation until 24 hours before departure
Refund	90% of approved amount; 10% academic penalty
Agency commission	5% of fare before tax; passenger price unchanged
Payment	Simulated; no PAN, CVV, or real payment provider
Demonstration data	Synthetic only

1.3 Scope exclusions

Round trips in one reservation, more than one connection, partial cancellations, itinerary changes, check-in, visual seat selection, baggage, meals, loyalty, promotional coupons, codeshare, notifications, real payments, multiple currencies, passenger age categories, and production authentication are outside the MVP. A return journey can be represented by a second one-way reservation. Flight coupons associated with issued tickets remain in scope.

2. Functional requirements

Priorities use MoSCoW. All requirements are approved as the final baseline for this implementation.

ID	Requirement	Actor	Priority
FR-001	Search one-way itineraries by origin, destination, local date, passenger count, and cabin.	Passenger, agency agent	Must
FR-002	Return direct itineraries and itineraries with at most one connection.	Passenger, agency agent	Must
FR-003	Enforce contiguous segments and a same-airport connection of 45 minutes to 6 hours.	System	Must
FR-004	Show schedules, local zones, cabin, minimum availability, total quote, and expiration.	Passenger, agency agent	Must

FR-005	Exclude expired holds from effective availability.	System	Must
FR-006	Reject reservations outside 180 days or inside the 60-minute sales cutoff.	Passenger, agency agent	Must
FR-007	Create one reservation for one itinerary and 1–9 passengers.	Passenger, agency agent	Must
FR-008	Hold every passenger/segment atomically or hold nothing.	System	Must
FR-009	Prevent negative inventory and duplicate active seat assignment on a flight leg.	System	Must
FR-010	Lock inventory rows in canonical order and recheck availability within the transaction.	System	Must
FR-011	Make reservation creation idempotent by actor, channel, and key for 24 hours.	Passenger, agency agent	Must
FR-012	Create a 60-minute PENDING_PAYMENT hold.	System	Must
FR-013	Assign one valid physical seat per passenger and segment automatically.	System	Must
FR-014	Expire unpaid reservations and release resources exactly once.	System	Must
FR-015	Retrieve a guest reservation using locator and passenger surname.	Passenger, agency agent	Must
FR-016	Record simulated payment attempts without cardholder data.	Payment simulator	Must
FR-017	Atomically confirm reservation and seats after timely approval.	Payment simulator	Must
FR-018	Issue one ticket per passenger and one coupon per itinerary segment.	System	Must
FR-019	Mark a declined reservation PAYMENT_FAILED and release resources once.	Payment simulator	Must
FR-020	Reject payment approval after hold expiration.	Payment simulator	Must
FR-021	Make payment processing idempotent by operation reference.	Payment simulator	Must
FR-022	Cancel a confirmed reservation until 24 hours before departure, void documents, release inventory, and record the refund.	Passenger, agency agent	Must

FR-023	Cancel a pending reservation without ticket or refund creation.	Passenger, agency agent	Must
FR-024	Make cancellation idempotent and preserve commercial history.	System	Must
FR-025	Reject partial cancellation and flight/date changes as unsupported.	Passenger, agency agent	Must
FR-026	Use the same inventory rules for agencies while recording agency and agent attribution.	Agency agent	Must
FR-027	Calculate a 5% agency commission before tax without changing passenger total.	System	Should
FR-028	Maintain airport, aircraft, seat, route, schedule, flight-instance, inventory, and fare catalogs.	Administrator	Must
FR-029	Enforce valid flight-instance state transitions.	Administrator	Should
FR-030	Reject aircraft substitution that cannot preserve active capacity and seat assignments.	Administrator	Must
FR-031	Return a read-only confirmed-passenger manifest without payment details.	Airport staff	Should
FR-032	Calculate price by route/cabin base, advance band, airport fee, and simulated tax.	System	Must
FR-033	Store and calculate COP amounts as fixed decimals with half-up two-decimal rounding.	System	Must
FR-034	Preserve an immutable price breakdown per passenger and segment.	System	Must
FR-035	Audit reservation, payment, ticket, and inventory state transitions with actor, time, reason, old/new state, and correlation ID.	System	Must
FR-036	Load approved revenue, refunds, cancellations, and occupancy snapshots into PostgreSQL OLAP.	ETL process	Must
FR-037	Distinguish voluntary cancellation from expiration and payment failure analytically.	ETL process, analyst	Must
FR-038	Prepare gross/net revenue, cancellation, and confirmed occupancy facts by relevant dimensions.	Analyst	Must
FR-039	Execute the analytical ETL automatically every hour using an activated Glue trigger.	ETL process	Must
FR-040	Publish selected OLTP and OLAP metadata in separate Glue Catalog databases.	ETL process, analyst	Must

The detailed acceptance criteria and origins are preserved in <docs/02-functional-requirements.md> .

3. Non-functional requirements

Each NFR defines a measurable condition and the business risk that motivates it. Production targets are distinct from what a low-cost Learner Lab can prove.

ID	Category	Verifiable target	Business risk and architectural consequence
NFR-01	Performance	At the assumed baseline dataset, flight search must achieve $p95 \leq 800$ ms for direct results and $p95 \leq 1.5$ s with one connection under 50 concurrent search clients. Reservation creation/query must achieve $p95 \leq 1.0$ s, excluding external payment time.	Slow search loses sales. Indexed relational searches and a thin API are used; future load testing must report p50/p95/p99.
NFR-02	Consistency/concurrency	With one remaining seat and 20 simultaneous independent PostgreSQL sessions, exactly one reservation succeeds, 19 receive INVENTORY_UNAVAILABLE, availability ends at zero, and no partial or duplicate seat assignment exists. Repeat 30 times for direct and two-segment bottlenecks.	Overselling creates operational and regulatory harm. PostgreSQL pessimistic locking, canonical lock order, atomic transactions, and constraints are mandatory.
NFR-03	Availability	Production target: 99.9% monthly availability for search and reservation APIs, excluding announced maintenance (≤ 43.8 minutes/month). Learner Lab is a Single-AZ demonstration and does not claim this SLA.	Peak downtime loses revenue. A production design would require Multi-AZ RDS, at least two API instances, health-checked load balancing, and automated recovery; these are excluded here for cost.
NFR-04	Recovery	Production OLTP RPO ≤ 5 minutes and RTO ≤ 60 minutes; OLAP RPO ≤ 1 hour and RTO ≤ 4 hours. Lab backups retain one day and teardown requires export/snapshot according to the runbook.	Reservation loss is more severe than rebuilding analytics. OLTP receives stricter recovery objectives; OLAP is reproducible from OLTP.
NFR-05	Security/privacy	RDS must remain private and encrypted; JDBC must require TLS; secrets must be absent from Git/logs; queries must be parameterized; API validation must reject malformed input; no PAN/CVV is stored; analytical	Passenger/payment exposure causes legal and reputational harm. Synthetic data is used. For the exercise, Colombian Law 1581/2012 principles guide PII minimization; PCI DSS

		tables must exclude direct passenger PII.	scope is avoided by not processing cards.
NFR-06	Scalability	The system must support 20 additional routes/year and the declared 5× search/3× booking peak without changing the domain model. Before 10× volume/frequency, search indexes and DB load are measured and full-snapshot ETL is replaced by auditable incremental staging/merge.	Database contention and full scans are the first likely bottlenecks. Stateless API scaling and isolated OLAP protect the transaction path.
NFR-07	Auditability	Every commercial state transition must be reconstructable by entity/correlation ID with actor, UTC time, reason, old/new state, and immutable financial snapshots. Academic evidence is retained for the course; a production retention period of five years is assumed pending legal policy.	Disputes and reconciliation require more than application logs. Audit rows and historical commercial records are not physically deleted.
NFR-08	Maintainability	Domain/application automated coverage must remain ≥90%; Ruff and strict mypy must pass; Alembic is the sole DDL authority; a clean PostgreSQL database must migrate from base to head reproducibly.	Divergent schemas and untested rules cause defects. CI-quality gates and versioned migrations enforce consistency.
NFR-09	Observability	Health endpoint, correlation IDs, sanitized errors, Glue run status/DPU seconds, ETL row counts and monetary deltas must be observable. A failed job or refresh older than 2 hours must raise an operational alert in production.	Silent pipeline failure produces stale or incorrect business data. Glue and analytics.etl_run provide independent evidence.
NFR-10	Analytical freshness	The Glue trigger runs at minute 0 every hour; each successful run records source_cutoff_at.	Commercial users need bounded staleness. Hourly freshness is balanced against Glue cost.
NFR-11	Analytical consistency	OLTP is read using REPEATABLE READ; all facts publish in one OLAP transaction; approved-payment and refund reconciliation deltas must be exactly COP 0.00.	Partial or mixed-cutoff facts lead to incorrect revenue. Atomic publication and reconciliation fail the run safely.

NFR-12	Analytical isolation/idempotency	ETL has concurrency 1, never writes OLTP, and rerunning the same <code>run_id</code> /snapshot cannot duplicate facts.	Analytics must not degrade or corrupt the booking system. Separate RDS instances and transaction-scoped foreign tables enforce the boundary.
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4. Entity-relationship model

4.1 Modeling decisions

The model uses **Crow's Foot notation** and is normalized for transactional integrity. A recurring `ScheduledFlight` contains ordered `ScheduledLeg` records; a dated `FlightInstance` contains operational `FlightLegInstance` records. A reservation owns one ordered itinerary and may contain multiple passengers. `ReservationItem` is the passenger × segment intersection and stores the immutable fare and active/historical seat state.

Authoritative inventory is one row per flight-leg instance and cabin. It stores capacity, held, and confirmed counts, constrained by `capacity >= held + confirmed`. Booking locks all required inventory rows in a canonical order. A partial unique PostgreSQL index prevents two active assignments of the same physical seat on the same leg while allowing historical released assignments.

```
erDiagram
    AIRPORT ||--o{ ROUTE : defines
    ROUTE ||--o{ SCHEDULED_FLIGHT : supports
    SCHEDULED_FLIGHT ||--|{ SCHEDULED_LEG : contains
    SCHEDULED_FLIGHT ||--o{ FLIGHT_INSTANCE : occurs_as
    FLIGHT_INSTANCE ||--|{ FLIGHT_LEG_INSTANCE : contains
    AIRCRAFT_TYPE ||--o{ AIRCRAFT : classifies
    AIRCRAFT ||--|{ SEAT : contains
    AIRCRAFT ||--o{ FLIGHT_INSTANCE : operates
    FLIGHT_LEG_INSTANCE ||--|{ INVENTORY : controls
    RESERVATION ||--|| ITINERARY : owns
    ITINERARY ||--|{ ITINERARY_SEGMENT : orders
    FLIGHT_LEG_INSTANCE ||--o{ ITINERARY_SEGMENT : references
    RESERVATION ||--|{ PASSENGER : contains
    PASSENGER ||--|{ RESERVATION_ITEM : travels_on
    FLIGHT_LEG_INSTANCE ||--o{ RESERVATION_ITEM : assigns
    SEAT ||--o{ RESERVATION_ITEM : allocated_as
    RESERVATION ||--o{ PAYMENT : attempts
    PAYMENT ||--o| REFUND : reverses
    PASSENGER ||--o| TICKET : receives
    TICKET ||--|{ COUPON : contains
    RESERVATION ||--o{ AUDIT_EVENT : records
    AGENCY ||--o{ AGENT : employs
    AGENCY ||--o{ RESERVATION : originates
```

Editable source: [diagramas/airline-oltp-erd.drawio](https://diagramas.airline-oltp-erd.drawio) .

4.2 Brief data dictionary

Entity	Key information and purpose
airports	IATA code PK, name, IANA time zone; public airport catalog.
cabins	Cabin code PK, order, active flag.
aircraft_types	Type identifier and cabin capacities.
aircraft	Tail/equipment instance referencing an aircraft type.
seats	Physical aircraft seat; unique label per aircraft and assigned cabin.
routes	Directed origin/destination airport pair and base commercial context.
scheduled_flights	Recurring commercial flight number and schedule.
scheduled_legs	Ordered legs of a recurring flight with local schedule and base fare.
flight_instances	Dated execution of a scheduled flight, aircraft, and operational state.
flight_leg_instances	Ordered dated leg with UTC departure/arrival and sellable state.
inventories	Authoritative capacity, held, and confirmed counts per leg/cabin.
reservations	Locator, state, channel/actor, hold expiry, total, commission, idempotency snapshot.
passengers	Synthetic given name and surname belonging to a reservation.
itineraries	One-to-one reservation itinerary container.
itinerary_segments	Ordered itinerary segments referencing flight-leg instances.
reservation_items	Passenger-segment seat assignment and immutable fare/tax/fee snapshot.
payments	Simulated attempt, COP amount, state, and unique operation reference; no card data.
refunds	Simulated refund linked to the approved payment.
tickets	Immutable ticket number/state per confirmed passenger.
coupons	One ticket coupon per passenger segment.
audit_events	Actor, correlation, entity, reason, previous/new state, UTC timestamp.
agencies / agents	Active partner agency and its authorized agent identities.
demo_identities	Controlled role fixtures for non-production authorization demonstrations.
fare_rules	Versioned pricing multipliers and fixed fare policy.
operational_settings	Hold, quote, sales, cancellation, tax, and refund policy values.
idempotency_records	Authoritative key scope, payload hash, response reference, and expiry.

Full column-level definitions are in [docs/data-dictionary.md](#) . Alembic migrations are the authoritative physical schema.

5. Backend architecture

5.1 Decision

The backend is a **modular monolith** implemented with FastAPI, SQLAlchemy, and PostgreSQL. Search, reservation, payment, ticketing, cancellation, audit, administration, and manifests share one deployable process but remain separated into API, application, domain, and persistence modules.

Microservices were rejected for the MVP because seat inventory, reservation, payment, and ticket effects require a single strong transaction. Splitting them would introduce distributed transactions, compensation, duplicate policy, and additional Academy cost without satisfying a present scale requirement.

5.2 Concurrency and transaction design

```
sequenceDiagram
    participant C as Client
    participant A as FastAPI
    participant S as BookingService
    participant P as PostgreSQL OLTP
    C->>A: POST /reservations + Idempotency-Key
    A->>S: validated command and actor
    S->>P: BEGIN
    S->>P: advisory lock for idempotency scope
    S->>P: SELECT inventories FOR UPDATE in canonical order
    S->>P: recheck capacity and assign seats
    alt all segments available
        S->>P: insert reservation/items/audit
        S->>P: COMMIT
        A-->>C: 201 PENDING_PAYMENT
    else any segment unavailable
        S->>P: ROLLBACK
        A-->>C: 409 INVENTORY_UNAVAILABLE
    end
end
```

This implements NFR-02 and FR-008-011. Canonical (leg, cabin) lock order reduces deadlock risk. Database CHECK, UNIQUE, foreign-key, and partial unique constraints provide a second line of defense.

5.3 Payment and agency exposure

Payment is simulated through the same backend; it accepts only an approval boolean and synthetic operation reference. Approval, reservation confirmation, inventory transition, ticket/coupon issuance, and audit occur atomically. The same versioned /api/v1 serves direct and agency channels because inventory rules are identical. Agency/agent attribution and authorization are explicit, avoiding duplicated booking logic. A future B2B gateway could expose the same application service with production authentication and rate limits.

5.4 Relationship to analytics

The API receives only OLTP credentials and never queries OLAP. Glue extracts a consistent OLTP snapshot and loads the separate analytical RDS. Analytical failure therefore does not block booking, satisfying NFR-12.

6. Frontend architecture

The frontend is a dependency-free, static single-page client using HTML, CSS, and JavaScript, served same-origin by FastAPI. This minimizes build/runtime complexity and network round trips for the academic MVP while supporting the search-to-cancellation workflow.

It implements direct and agency modes in one interface because both consume the same domain rules. Runtime data is never mocked. If availability changes between search and booking, the API returns a structured `409 INVENTORY_UNAVAILABLE` ; the UI shows the conflict and requires a fresh search. This is request/response feedback rather than WebSocket push, appropriate to the MVP and consistent with database-authoritative availability.

The interface supports English labels and COP only. Internationalization and multiple currencies are explicit future extensions: presentation strings and currency formatting would be separated from the API's immutable COP amounts; exchange-rate conversion would never alter stored transaction values.

The browser client consumes:

- `GET /api/v1/flights/search ;`
- `POST /api/v1/reservations ;`
- `GET /api/v1/reservations/{locator} ;`
- `POST /api/v1/reservations/{id}/payments ;`
- `POST /api/v1/reservations/{id}/cancel ;`
- `GET /api/v1/reservations/{id}/tickets .`

7. First backend implementation

7.1 Implemented critical operations

FastAPI exposes the required search, reservation creation, and reservation query operations plus payment, cancellation, tickets, manifests, health, and controlled administration. PostgreSQL implements the complete ER model and is managed exclusively with Alembic migrations `0001` through `0008` .

7.2 Concrete concurrency proof

The real PostgreSQL concurrency suite exercises two parametrized cases:

- direct itinerary with one remaining seat;
- two-segment itinerary whose second segment is the bottleneck.

Each case runs 30 iterations of 20 independent concurrent database sessions: 1,200 reservation attempts in total. Every iteration asserts exactly one winner, 19 `INVENTORY_UNAVAILABLE` conflicts, zero remaining availability, no negative counter, no duplicate active seat, and no partial reservation.

Recorded evidence:

```
AIRLINE_CONCURRENCY_ITERATIONS=30 python3 -m pytest -q -m concurrency
2 passed
```

Additional recorded quality evidence includes 19 unit/API tests, 21 non-concurrency PostgreSQL integration tests, four analytical scenarios, Ruff success, strict mypy success, and $\geq 90\%$ scoped domain/application coverage. A fresh audit on September 10 also produced 23 passing unit/API/analytics tests, 27 passing frontend tests, and successful positive/negative infrastructure validation.

7.3 Execution guide

Local PostgreSQL and application:

```
cd backend
docker compose up -d
python -m pip install -e '.[dev]'
alembic upgrade head
python -m airline_core.persistence.seed_cli --include-demo
uvicorn airline_core.main:app --host 0.0.0.0 --port 8000
```

Tests:

```
cd backend
python -m pytest -q
AIRLINE_CONCURRENCY_ITERATIONS=30 python -m pytest -q -m concurrency
cd ../frontend && node test.js
cd .. && ./infra/validate.sh && ./infra/validate-negative.sh
```

AWS deployment is performed with `deploy/create-stack.sh`, `deploy/ec2-run.sh`, and `deploy/update-analytics-stack.sh`; secrets are entered interactively or passed at runtime and are not versioned.

8. AWS analytical implementation

8.1 Deployed architecture

```
flowchart LR
    U["U[Passenger / Agency Browser] -->|HTTP demo :8000| E[EC2 t3.micro\nFastAPI + static SPA]"]
    E -->|private PostgreSQL/TLS| O["O[(RDS PostgreSQL\nairline_oltp\ndb.t3.micro)]"]
    O --> T["T[Glue hourly trigger] --> G[AWS Glue 4.0 ETL\n2 x G.1X, concurrency 1]"]
    T --> S3["S3[(S3 versioned\nscript and temporary files)] --> G"]
    G -->|JDBC/TLS, read-only source| O
    G -->|atomic refresh| A["A[(RDS PostgreSQL\nairline_analytics\ndb.t3.micro)]"]
    C1["C1[OLTP crawler] --> O"]
    C2["C2[OLAP crawler] --> A"]
    C1 --> D1["D1[Glue Catalog\nairline_oltp]"]
    C2 --> D2["D2[Glue Catalog\nairline_analytics]"]
```

All resources are in `us-east-1`. EC2 is in a public subnet for the demo. Both RDS databases are private, encrypted, Single-AZ, and placed in a two-subnet DB subnet group. Glue uses private ENIs and security-group rules. An S3 gateway endpoint avoids a permanent NAT Gateway. The editable detailed diagram is [diagramas/arquitectura-aws.drawio](https://diagramas.arquitectura-aws.drawio).

8.2 Analytical model and ETL

The OLAP schema is dimensional rather than a copy of the normalized OLTP schema. It contains:

- dimensions: date, route, flight, cabin, channel, agency, and fare band;
- facts: reservation, passenger-segment sales, and leg-occupancy snapshot;
- bridge: reservation-to-route;

- control: ETL run and watermark.

This grain supports occupancy by route/cabin, gross and net revenue by fare, and cancellation patterns by route/date/channel/advance band. Direct passenger PII, actor IDs, agent IDs, documents, and idempotency keys are excluded.

The Glue job is the orchestrator; the versioned PostgreSQL function

`analytics.refresh_warehouse(run_id, snapshot_at)` performs set-based transformation. The job opens transaction-scoped `postgres_fdw` foreign tables for 12 selected OLTP sources, calls the function in `REPEATABLE READ`, drops the foreign server before commit, and records success/failure separately. A full snapshot plus UPSERT captures late payment, cancellation, and refund changes; a `created_at`-only bookmark would miss these updates.

8.3 AWS service justification

Service/component	Alternative	Supporting requirement	Well-Architected justification
EC2 t3.micro API	Lambda/ECS	FR-001-035; NFR-01/08	Cost optimization: small on-demand host. Operational excellence: Dockerized reproducible runtime. Performance efficiency: adequate measured CPU for MVP.
RDS PostgreSQL OLTP	PostgreSQL on EC2	FR-007-035; NFR-02/04/05	Reliability: managed backups and transactional durability. Security: private encrypted DB. Operational excellence: managed engine.
Separate RDS PostgreSQL OLAP	Shared schema or Redshift	FR-036-038; NFR-11/12	Reliability/performance: analytics cannot contend for OLTP compute. Cost: minimum Single-AZ class instead of a warehouse.
Glue 4.0 Spark job	Lambda or EC2 cron	FR-036/039; NFR-10/12	Operational excellence: managed scheduling and run history. Performance: set-based refresh. Cost/sustainability: two-worker minimum and disabled outside needed windows.
Glue trigger	cron on EC2	FR-039; NFR-10	Reliability/operations: declarative hourly schedule independent of API process.
Two JDBC connections/crawlers	Manual metadata	FR-040	Operational excellence: reproducible schema discovery. Reliability: separate OLTP/OLAP catalogs. Crawlers run only after schema changes.

Glue Data Catalog	Handwritten data dictionary only	FR-040	Operational excellence: centralized queryable metadata; usage is below free thresholds.
Versioned encrypted S3 bucket	Script stored on EC2	FR-039; NFR-08/09	Reliability/security: recoverable ETL artifact, blocked public access, encryption.
S3 gateway endpoint	NAT Gateway	NFR-05/12	Security/cost/sustainability: private S3 access without NAT hourly/processing cost.
VPC/private subnets/security groups/TLS	Public RDS	NFR-05/12	Security: least network exposure. Reliability: controlled connectivity.
Existing Academy LabRole	Dedicated least-privilege role	Academy constraint; FR-039/040	Operational excellence: only viable lab integration. This is a documented security exception ; production requires dedicated roles.

8.4 Live evidence

Read-only verification on September 10, 2026 established:

- CloudFormation stack `airline-demo` : `UPDATE_COMPLETE` ;
- EC2 API: running and `/api/v1/health` returned `status=ok` ;
- OLTP and OLAP RDS: PostgreSQL 16.10, `db.t3.micro` , 20 GB gp3, encrypted, private, Single-AZ;
- Glue ETL: two successful executions; 79 s/159 DPU-s and 131 s/262 DPU-s;
- trigger `airline-analytics-etl-hourly` : `ACTIVATED` , `cron(0 * * * ? *)` ;
- crawlers: both `SUCCEEDED` ;
- Catalog `airline_oltp` : 12 authorized source tables;
- Catalog `airline_analytics` : 13 dimensional/control tables;
- analytical load evidence: 1,406 rows extracted/loaded with `approved_delta=0.00` and `refund_delta=0.00` .

9. Monthly cost projection

9.1 Assumptions

Prices are public list prices for `us-east-1` , 730 hours/month, with no Academy credits, Free Tier, taxes, extraordinary transfer, or human labor. Glue costs USD 0.44/DPU-hour. The scheduled observed run consumed 262 DPU-seconds. Crawlers use 2 DPU and have a 10-minute billing minimum. S3 and Catalog usage remain negligible at current scale.

Component	Calculation	USD/month
EC2 t3.micro	730×0.0104	7.59
Public IPv4	730×0.005	3.65
EBS gp3, 8 GB	8×0.08	0.64

OLTP RDS compute	730×0.018	13.14
OLTP RDS gp3, 20 GB	20×0.115	2.30
Transactional application		27.32
OLAP RDS compute	730×0.018	13.14
OLAP RDS gp3, 20 GB	20×0.115	2.30
Hourly Glue ETL	$730 \times 262/3,600 \times 0.44$	23.38
Two weekly crawlers	$4.33 \times 2 \times 2 \times 10/60 \times 0.44$	1.27
Catalog, S3, current logs	below free/negligible thresholds	<0.01
Incremental analytics		40.09
Total run rate		67.41

9.2 Growth and frequency scenarios

Scenario	Assumption	Total/month	Architectural response
Current	Hourly, 262 DPU-s/run	USD 67.41	Measure at least 30 runs and retain hourly only if required.
Daily refresh	30 runs/month	USD 44.99	Best low-cost laboratory configuration; 24-hour freshness.
Every 4 hours	183 runs/month	USD 49.89	Middle ground between cost and freshness.
Five-minute hourly ETL	2 DPU for 5 minutes/run	USD 97.57	Optimize SQL and incremental extraction before scaling workers.
10× ETL duration	2,620 DPU-s/run	USD 277.82	Replace full snapshots with auditable CDC/staging/merge; measure RDS and cross-AZ transfer.
10× frequency	7,300 equivalent current runs	approximately USD 277.82	Do not micro-batch Spark at this scale; group changes or use an event/incremental design.

The frequency and duration 10× cases have the same Glue compute multiplication under this simplified model. Volume growth can additionally require larger RDS classes/storage and increase cross-AZ transfer, so USD 277.82 is not an upper bound. Sustainability and cost optimization both favor processing only changed rows and disabling scheduled compute when no consumer needs fresh data.

10. Design-to-implementation deviations

Initial idea or ambiguity	Final implementation	Reason and consequence
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Analytical schema initially co-located with OLTP	Separate OLAP RDS and migration/bootstrap path	The assignment calls for a second analytical database and isolation prevents analytical contention. Cost increases by USD 15.44/month.
ETL transformation could live in Python/Spark	Glue orchestrates a versioned PostgreSQL set-based function	One transformation contract serves local and AWS execution, avoids unavailable Glue Python packages, and permits atomic reconciliation.
Simple reservation-column idempotency	Authoritative <code>idempotency_records</code> plus advisory lock; reservation fields remain audit snapshots	Correctly enforces the 24-hour scope and detects same-key/different-payload conflicts under concurrency.
Agency identifiers were plain attributes	Foreign keys to active agency/agent catalogs and controlled demo identities	Prevents forged or incoherent agency attribution.
Frontend UUID generation assumed secure context	Web Crypto UUID fallback for public HTTP demo	<code>crypto.randomUUID()</code> is commonly unavailable outside HTTPS; the fallback preserves idempotency in Academy demo access.
Public EC2 address could change	A manually associated project Elastic IP stabilizes the demo	Improves demonstrability but adds USD 3.65/month and exists outside the CloudFormation lifecycle.
Production-style least-privilege IAM	Existing broad <code>LabRole</code>	Academy forbids project IAM changes. This is not a production recommendation.

11. Traceability summary

Concern	Requirement	Design/implementation	Evidence
No overselling	FR-008-010; NFR-02	canonical FOR UPDATE, constraints, atomic service	20 clients × 30 repetitions for direct and multi-leg cases
Multi-leg consistency	FR-002/003/007/008	itinerary/segments and all-or-nothing inventory	integration rollback and ticket tests
Safe retries	FR-011/021/024	idempotency records, hashes, unique operation references	reservation/payment/cancellation replay tests
Revenue correctness	FR-032-038; NFR-11	immutable price facts and reconciled warehouse refresh	zero COP approved/refund deltas
OLTP isolation	FR-036; NFR-12	separate RDS, read-only extraction path	API remains healthy during/after analytical deployment

Metadata visibility	FR-040	two connections, crawlers, Catalog databases	12 OLTP and 13 OLAP live tables
Freshness	FR-039; NFR-10	activated hourly Glue trigger	scheduled successful job run
Privacy	NFR-05	private encrypted RDS, TLS, PII-free OLAP	IaC validation and schema contract tests

12. Limitations and production evolution

This is an educational MVP, not a production airline platform. The demo uses HTTP, a public EC2 port, static demonstration identities, Single-AZ resources, a broad Academy role, one API host, no WAF/load balancer, and one-day backup retention. The EC2 root volume is currently unencrypted. These choices are budget/environment compromises and cannot satisfy the stated production availability and recovery targets.

A production evolution would introduce TLS termination and DNS, real identity and role-based authorization, Secrets Manager, least-privilege IAM, Multi-AZ RDS, multiple stateless API tasks behind a load balancer, WAF/rate limiting, centralized metrics/alerts, tested point-in-time recovery, encrypted compute volumes, data-retention governance, incremental change capture, and controlled deployment pipelines.

13. Prompt log appendix

AI assistance was used as permitted by the assignment. The student reviewed the resulting decisions and must be able to defend them. No credentials or private access tokens are included.

Stage	Main prompt intent	Refinement/correction by the team	Resulting artifact
Planning	Analyze the PDF, create a rigorous question set, account for Learner Lab IAM and USD 45 remaining budget.	The team confirmed no special constraint, one-person team, English submission, RDS preference, simple frontend, and a maximum USD 15 additional spend.	Project definition and Academy register.
Functional definition	Make requirements realistic for an airline while remaining defensible.	Scope was constrained to one-way/two-segment travel, synthetic payments, whole-reservation cancellation, COP, automated seats, and shared agency rules.	FR-001–040 and explicit exclusions

Core engine	Build the reservation engine first with strong automated evidence.	PostgreSQL—not mocks/SQLite—was required for locking evidence; pricing/state/idempotency invariants were made testable.	Domain/application service and concurrency tests.
Database hardening	Make Alembic the only DDL authority and prove repeatable bootstrap/migrations.	Required catalogs were separated from synthetic demo fixtures; legacy migration and constraint tests were added.	Migrations, bootstrap, schema tests, data dictionary.
MVP completion	Add API, minimal frontend, deployment, and simulations without analytics yet.	Parallel work was integrated only after domain rules and PostgreSQL behavior were stable.	FastAPI, SPA, Docker/AWS scripts, simulation evidence.
Analytical MVP	Add data engineering but not dashboards/interpretation.	Physical OLTP/OLAP isolation replaced the earlier co-located idea; PII was excluded and reconciliation made mandatory.	Dimensional model, ETL, separate RDS bootstrap, analytical tests.
Scheduler	Execute ETL automatically every hour.	Concurrency was limited to one; the trigger is activated only after the script is uploaded.	Glue scheduled trigger and runbook.
Diagrams/documentation	Produce minimal ETL, data-flow, and integrated AWS diagrams.	Editable Draw.io sources were centralized and expanded to reflect separate RDS instances, catalogs, private networking, and failure controls.	diagrams/ and analytical design documents.
Cost audit	Rigorously analyze repository and live AWS costs.	Static five-minute Glue assumptions were replaced by observed DPU-seconds; crawler minimum billing and manually managed EIP were included.	docs/cost-analysis.md and section 9 of this report.
Final compliance audit	Compare every PDF deliverable with code and deployed AWS.	Missing formal NFR decisions were closed consistently with implemented behavior;	docs/midterm-glue-analysis.md and this final report

		contradictions were identified for correction.	
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The fuller historical prompt record is retained in [Prompts.md](#) .

14. References and repository evidence

- Assignment: `parcial12026-2-EN.docx.pdf` .
- Functional baseline: [docs/02-functional-requirements.md](#) .
- Detailed dictionary: [docs/data-dictionary.md](#) .
- Transaction design and traceability: [docs/database-design.md](#) , [docs/core-engine-traceability.md](#) .
- Architecture decisions: [docs/adr/](#) .
- Analytical design: [docs/analytics-detailed-design.md](#) , [docs/analytics-model.md](#) .
- Deployment and operation: [docs/deployment/README.md](#) , [docs/deployment/analytics-runbook.md](#) .
- Test/deployment evidence: [docs/evidence/](#) .
- Cost audit: [docs/cost-analysis.md](#) .
- AWS Glue pricing: <https://aws.amazon.com/glue/pricing/>.
- AWS public IPv4 pricing: <https://aws.amazon.com/vpc/pricing/>.
- AWS Well-Architected pillars: <https://docs.aws.amazon.com/wellarchitected/latest/framework/the-pillars-of-the-framework.html>.

Submission checklist

Before uploading, the student must replace the source-revision placeholder with the GitHub URL and immutable commit/tag, export missing diagram previews, rerun the full PostgreSQL test suite, correct stale standalone AWS evidence, and ensure the deployed resources are either intentionally retained for the oral defense or safely removed to protect the Academy budget.